

# Data Report Write Up

Emma DiLauro

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```
load("G:\\2018-2019\\Boot Camp\\Canned Data Report - Boot Camp\\mturkstress.rda")
#load in dataset from shared drive
```

## Variables

How many subjects were in each of the four conditions of the experiment?

```
conditions<- table(mturkstress$treatmentgroup)
#creates basic table of the variable

rownames(conditions) <- c("Control", "Life", "Pol", "Brighten")
# creates names for the rows of the table instead of having 0,1 etc

conditions<- xtable(conditions, caption="Subjects in Each Treatment Group")
# creates the nicely formatted table and assigns this to name conditions

print(conditions)
```

```
## % latex table generated in R 3.5.2 by xtable 1.8-3 package
## % Wed Jan 30 16:29:18 2019
## \begin{table}[ht]
## \centering
## \begin{tabular}{rr}
## \hline
## & V1 \\
## \hline
## Control & 448 \\
## Life & 445 \\
## Pol & 440 \\
## Brighten & 446 \\
## \hline
## \end{tabular}
## \caption{Subjects in Each Treatment Group}
## \end{table}
```

*#prints Latex code, then copy paste it into next line so that it turns into the table when knitted to a*

% latex table generated in R 3.5.2 by xtable 1.8-3 package % Mon Jan 28 22:32:16 2019

|          | x   |
|----------|-----|
| control  | 448 |
| life     | 445 |
| pol      | 440 |
| brighten | 446 |

Table 1: Subjects in Each Treatment Group

The control group contained 448 subjects, the group treated with stress about their life had 445 subjects, the group treated with political stress had 440 subjects, and the group treated with things that brightened their day had 446 participants.

What is the mean of the Social Interaction Anxiety Scale variable? (saisscale)

```
summary(mturkstress$saisscale)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.     NA's
## 20.00   34.25   47.00   48.35   61.00   100.00      92
```

*#outputs a summary of important values/markers (min/max, quartiles, mean)*

The mean of the Social Interaction Anxiety Scale variable is 48.35.

Create a new variable called “partystrength” that takes these values: 0 if the respondent is a true Independent, 1 if the respondent leans toward the Democrats or Republicans, 2 if the respondent is a weak Democrat or Republican, and 3 if the respondent is a strong Democrat or Republican. Make a table of the distribution of your new variable.

```
mturkstress$partystrength <- NA
# creates new variable that is empty

#0 if the respondent is a true Independent, 1 if the respondent leans toward the Democrats or Republicans
mturkstress$partystrength <- 0
#default value is 0 for the variable

mturkstress$partystrength[which(mturkstress$part2 == "1" | mturkstress$part2 == "2")] <- 1
#if response was 1 or 2, assign value to 1 in the new variable

mturkstress$partystrength[which(mturkstress$part3 == "2")] <- 2
#if response was 2, assign value to 2 in the new variable

mturkstress$partystrength[which(mturkstress$part3 == "1")] <- 3
#if response was 1, assign value to 3 in the new variable

party <- table(mturkstress$partystrength)
#creates basic table of the variable

rownames(party) <- c("True Independent", "Leans Democrat/Republican", "Weak Democrat/Republican", "Strong Democrat/Republican")
# creates names for the rows of the table instead of having 0,1 etc

party <- xtable(party, caption="Party Strength of Respondents")
# creates the nicely formatted table and assigns this to name party

print(party)
```

```
## % latex table generated in R 3.5.2 by xtable 1.8-3 package
## % Wed Jan 30 16:29:18 2019
## \begin{table}[ht]
## \centering
## \begin{tabular}{rr}
## \hline
## & V1 \\
## \hline
## True Independent & 285 \\
## Leans Democrat/Republican & 461 \\
## Weak Democrat/Republican & 577 \\
## Strong Democrat/Republican & 603 \\
## \hline
## \end{tabular}
## \caption{Party Strength of Respondents}
```

```
## \end{table}
```

*#prints Latex code, then copy paste it into next line so that it turns into the table when knitted to a*

% latex table generated in R 3.5.2 by xtable 1.8-3 package % Tue Jan 29 13:21:14 2019

|                            | V1  |
|----------------------------|-----|
| True Independent           | 285 |
| Leans Democrat/Republican  | 461 |
| Weak Democrat/Republican   | 577 |
| Strong Democrat/Republican | 603 |

Table 2: Party Strength of Respondents

285 people are True Independents, 461 lean either towards Democrats or Republicans, 577 people are weak Democrats or Republicans, and 603 are strong Democrats or Republicans.

Create an index variable (a variable that aggregates information from multiple variables) for the “political engagement scale” (polengage1-polengage6). First, recode any of the six variables that are reverse coded. Then, create an additive index (where you add up a respondent’s values on each of the six variables to get a total score).

```
#NEW VARIABLES WHERE 5 IS MOST ENGAGED (CHANGE VARIABLES 2, 4, 5)
polengage2_new <- (6 - mturkstress$polengage2)
#subtracts 6 from answer to recode variable in reverse order

polengage4_new <- (6- mturkstress$polengage4)
#subtracts 6 from answer to recode variable in reverse order

polengage5_new <- (6-mturkstress$polengage5)
#subtracts 6 from answer to recode variable in reverse order

polengage_index <- (mturkstress$polengage1 + mturkstress$polengage2_new + mturkstress$polengage3 + mturkstress$polengage4_new + mturkstress$polengage5_new)
#adds score from each question to create additive index
```

Must recode variables 2, 4, and 5 so that “agree” represents more politically engaged and index variable can be properly created.

Identify three variables that interest you in the dataset. Create separate tables that show the distribution of values for each variable.

```
avoid <- table(mturkstress$discussavoid)
#creates basic table of the variable
rownames(avoid) <- c("Avoid", "Enjoy", "Somewhere in between")
# creates names for the rows of the table instead of having 0,1 etc
avoid<- xtable(avoid, caption= "Avoidance of Political Discussions")
# creates the nicely formatted table and assigns this to name avoid

community <- table(mturkstress$comgroup)
#creates basic table of the variable
rownames(community) <- c("yes, active member", "yes, but not active", "no, not a member")
# creates names for the rows of the table instead of having 0,1 etc
```

```

community <- xtable(community, caption= "Involvement in Community Groups")
# creates the nicely formatted table and assigns this to name community

vote<- table(mturkstress$vote2008)
#creates basic table of the variable
rownames(vote) <- c("I did not vote", "thought about voting but didn't", "usually vote, but didn't", "su
# creates names for the rows of the table instead of having 0,1 etc
vote <- xtable(vote, caption= "Vote in 2008")
# creates the nicely formatted table and assigns this to name vote

print(avoid)

## % latex table generated in R 3.5.2 by xtable 1.8-3 package
## % Wed Jan 30 16:29:18 2019
## \begin{table}[ht]
## \centering
## \begin{tabular}{rr}
## \hline
## & V1 \\
## \hline
## Avoid & 569 \\
## Enjoy & 343 \\
## Somewhere in between & 943 \\
## \hline
## \end{tabular}
## \caption{Avoidance of Political Discussions}
## \end{table}

#prints Latex code, then copy paste it into next line so that it turns into the table when knitted to a
print(community)

## % latex table generated in R 3.5.2 by xtable 1.8-3 package
## % Wed Jan 30 16:29:18 2019
## \begin{table}[ht]
## \centering
## \begin{tabular}{rr}
## \hline
## & V1 \\
## \hline
## yes, active member & 289 \\
## yes, but not active & 218 \\
## no, not a member & 1322 \\
## \hline
## \end{tabular}
## \caption{Involvement in Community Groups}
## \end{table}

#prints Latex code, then copy paste it into next line so that it turns into the table when knitted to a
print(vote)

## % latex table generated in R 3.5.2 by xtable 1.8-3 package
## % Wed Jan 30 16:29:18 2019
## \begin{table}[ht]

```

```
## \centering
## \begin{tabular}{rr}
##   \hline
##   & V1 \\
##   \hline
## I did not vote & 517 \\
##   thought about voting but didn't & 159 \\
##   usually vote, but didn't & 68 \\
##   sure I voted & 1053 \\
##   \hline
## \end{tabular}
## \caption{Vote in 2008}
## \end{table}
```

*#prints Latex code, then copy paste it into next line so that it turns into the table when knitted to a*

Show in New WindowClear OutputExpand/Collapse Output % latex table generated in R 3.5.2 by xtable 1.8-3 package % Wed Jan 30 11:12:30 2019

|          | V1  |
|----------|-----|
| Control  | 448 |
| Life     | 445 |
| Pol      | 440 |
| Brighten | 446 |

Table 3: Subjects in Each Treatment Group

Show in New WindowClear OutputExpand/Collapse Output Min. 1st Qu. Median Mean 3rd Qu. Max. NA's  
20.00 34.25 47.00 48.35 61.00 100.00 92 Show in New WindowClear OutputExpand/Collapse Output % latex  
table generated in R 3.5.2 by xtable 1.8-3 package % Wed Jan 30 11:12:31 2019

|                            | V1  |
|----------------------------|-----|
| True Independent           | 285 |
| Leans Democrat/Republican  | 461 |
| Weak Democrat/Republican   | 577 |
| Strong Democrat/Republican | 603 |

Table 4: Party Strength of Respondents

Show in New WindowClear OutputExpand/Collapse Output % latex table generated in R 3.5.2 by xtable 1.8-3 package % Wed Jan 30 11:15:25 2019

|                      | V1  |
|----------------------|-----|
| Avoid                | 569 |
| Enjoy                | 343 |
| Somewhere in between | 943 |

Table 5: Avoidance of Political Discussions

% latex table generated in R 3.5.2 by xtable 1.8-3 package % Wed Jan 30 11:15:25 2019

% latex table generated in R 3.5.2 by xtable 1.8-3 package % Wed Jan 30 11:15:25 2019

The avoiding political discussions, community group involvement, and voting in the 2008 election variables all interested me. For avoiding political discussion, 569 people say they avoid the conversations, 343 enjoy them, and 943 are in between. For community groups, 289 people said they were an active member of one of the listed groups, 218 said they were, but not an active one, and 1322 people said they were not members

|                     | V1   |
|---------------------|------|
| yes, active member  | 289  |
| yes, but not active | 218  |
| no, not a member    | 1322 |

Table 6: Involvement in Community Groups

|                                 | V1   |
|---------------------------------|------|
| I did not vote                  | 517  |
| thought about voting but didn't | 159  |
| usually vote, but didn't        | 68   |
| sure I voted                    | 1053 |

Table 7: Vote in 2008

of any of the groups. Finally, for the 2008 election variable, 517 people said they didn't vote, 159 people reported that they thought about voting but ultimately didn't, 68 people who usually vote didn't in this election, and 1053 respondents voted.

**Identify two variables that you think might have a relationship with each other. What is your expectation about the nature of that relationship? Which variable would be the independent variable and which the dependent?**

I think that community group involvement and vote likeliness might have a relationship with each other. In one of my classes I read 'Making Democracy Work' by Putnam, and he explores this idea throughout his work. Community group involvement would be the independent variable, with vote likeliness as the dependent variable. I would expect, based on Putnam's work as well, that those individuals who stated they are active members of community groups will be more likely to vote.

## Sample

**What is the unit of analysis in this data set?**

The individual

**What proportions of respondents finished the survey?**

```
prop.table(table(mturkstress$finished))
```

```
##
##           0           1
## 0.0768432 0.9231568
```

*#creates table with calculates proportions*

92.3% of respondents finished the survey.

**What proportion of the sample is married? What proportion of the women in the sample is married?**

```
table(mturkstress$marital[which(mturkstress$marital == "2")])
```

```
##
##      2
```

```
## 602
# creates simple table of total married in sample

table(mturkstress$male[which(mturkstress$marital == "2")])
```

```
##
## 1 2
## 246 356
```

```
#creates simple table of men, women who are married

table(mturkstress$male)
```

```
##
## 1 2
## 991 932
```

```
#creates simple table of number of men, women
```

```
#can then use the values presented in the tables for calculations
```

991+ 932 = 1,923 - total respondents (602/1,923)\* 100 = 31% of respondents are married (356/932)\*100 = 38% of the women in the sample are married

Are the proportion of men who are married and the proportion of women who are married in the sample significantly different from one another? Use the `prop.test` command.

```
men_married <- length(mturkstress$male[which(mturkstress$male == "1" & mturkstress$marital == "2")])
#creates variable of number of men who are married

women_married <- length(mturkstress$male[which(mturkstress$male == "2" & mturkstress$marital == "2")])
#creates variable of number of women who are married

total_men <- length(mturkstress$male[which(mturkstress$male == "1")])
#creates variable of total men in sample

total_women <- length(mturkstress$male[which(mturkstress$male == "2")])
#creates variable of total women in sample

prop.test(x=c(men_married,women_married), n=c(total_men,total_women))
```

```
##
## 2-sample test for equality of proportions with continuity
## correction
##
## data: c(men_married, women_married) out of c(total_men, total_women)
## X-squared = 39.328, df = 1, p-value = 3.582e-10
## alternative hypothesis: two.sided
## 95 percent confidence interval:
## -0.17596852 -0.09151176
## sample estimates:
## prop 1 prop 2
## 0.2482341 0.3819742
```



```
#runs a proportions test with the values to see if there's a significant difference
```

Yes the difference in the proportion of men and women who are married is significant- since the p value is less than .05.

Pretend that the data we collected were a national probability sample. (They're not, but pretend.) First go to the American National Election Study website (<http://electionstudies.org/nesguide/gd-index.htm#2>) and find the total proportion of people in 2012 who were strong Democrats, weak Democrats, or leaned Democrat. Assume that there were 1,000 individuals in the ANES sample. Is the proportion of Democrats in the ANES significantly different from the equivalent proportion in our dataset? How does this affect our interpretation of our findings?

```
#ANES Data, not sure if used the right dataset? Liberal-Conservative Self-Identification 2012

leans_dem_anes <- 691
weak_dem_anes <- 890
strong_dem_anes <- 1156

total_dem_anes <- (leans_dem_anes + weak_dem_anes + strong_dem_anes)
total_respondents_anes<- 5871

#our data
strong_democrats_mturk <- length(mturkstress$part1[which(mturkstress$part1 == "1" & mturkstress$part3 == "1")])
#adds all respondents that answered democrat and strong

weak_dem_mturk <- length(mturkstress$part1[which(mturkstress$part1 == "1" & mturkstress$part3 == "2")])
#adds all respondents that answered democrat and weak

leans_dem_mturk <- length(mturkstress$part1[which(mturkstress$part1 == "3" & mturkstress$part2 == "1")])
#add all respondents that answered independent and leans democrat

total_dem_mturk <- (strong_democrats_mturk + weak_dem_mturk + leans_dem_mturk)
#adds all democrats

total_respondents_mturk <- 1923

prop.test(x=c(total_dem_anes,total_dem_mturk), n=c(total_respondents_anes, total_respondents_mturk))

##
## 2-sample test for equality of proportions with continuity
## correction
##
## data: c(total_dem_anes, total_dem_mturk) out of c(total_respondents_anes, total_respondents_mturk)
## X-squared = 145.63, df = 1, p-value < 2.2e-16
## alternative hypothesis: two.sided
## 95 percent confidence interval:
## -0.1843400 -0.1334105
## sample estimates:
## prop 1 prop 2
## 0.4661897 0.6250650
```

```
#runs test to see if proportions are significantly different
```

Not significantly different, prop.test tests for equality of the proportions, which is significant since p-value is below .05. Therefore, this shows us that our findings should be an accurate representation, since they aren't significantly different from the trusted ANES data.

## Hypothesis Testing

Did the political stress treatment cause respondents to report a lower likelihood of voting compared to respondents in the control group? First, be clear on what test you are running to assess this hypothesis. I will check your code to see if you implemented the test correctly. In your report, you should clearly explain your finding.

```
x <- mturkstress$voteslikely[which(mturkstress$treatmentgroup == "2")]  
#assigns all values for voteslikely where respondent was in pol stress group to object x  
  
y <- mturkstress$voteslikely[which(mturkstress$treatmentgroup == "0")]  
#assigns all values for voteslikely where respondent was in control group to object y  
  
t.test(x,y)
```

```
##  
## Welch Two Sample t-test  
##  
## data: x and y  
## t = -0.50572, df = 882.99, p-value = 0.6132  
## alternative hypothesis: true difference in means is not equal to 0  
## 95 percent confidence interval:  
## -5.86049 3.45911  
## sample estimates:  
## mean of x mean of y  
## 77.13636 78.33705
```

```
#runs t test to check for significance of effect of being in political stress group on voteslikely response
```

No the political stress treatment did not cause respondents to report a lower likelihood of voting because the test is not significant - the p value is greater than .05.

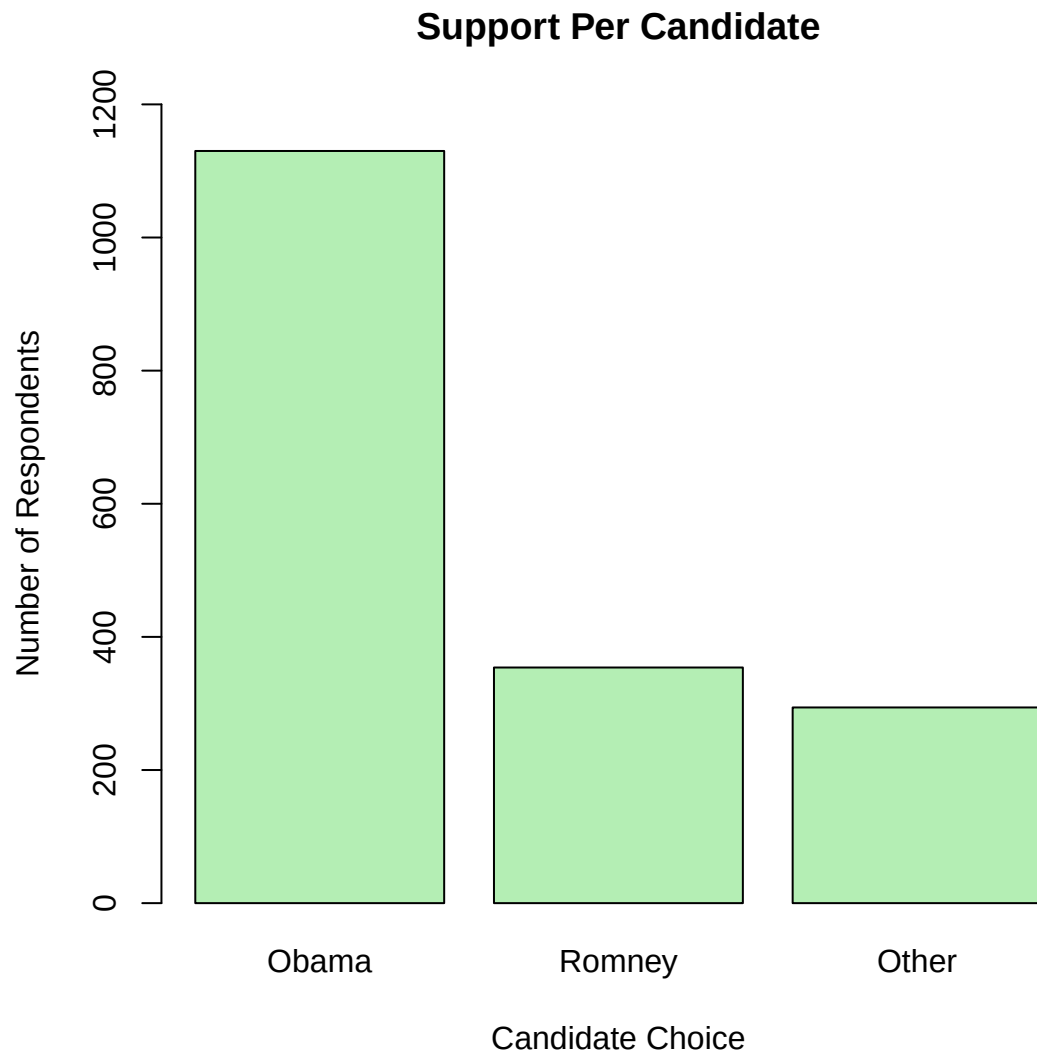
## Data Visualization

Note: For all graphs, label your axes and title your plot. For example code on how to do this, see the “Example Data Report” code.

Make a bar chart of the proportion of supporters for each candidate (votechoice).

```
support <- table(mturkstress$votechoice)  
#creates simple table of votechoice and assigns it to object support  
  
names(support) <- c("Obama", "Romney", "Other")  
#put names of candidates on barplot
```

```
barplot(support, col = "darkseagreen2", xlab= "Candidate Choice", ylab= "Number of Respondents", ylim=c
```

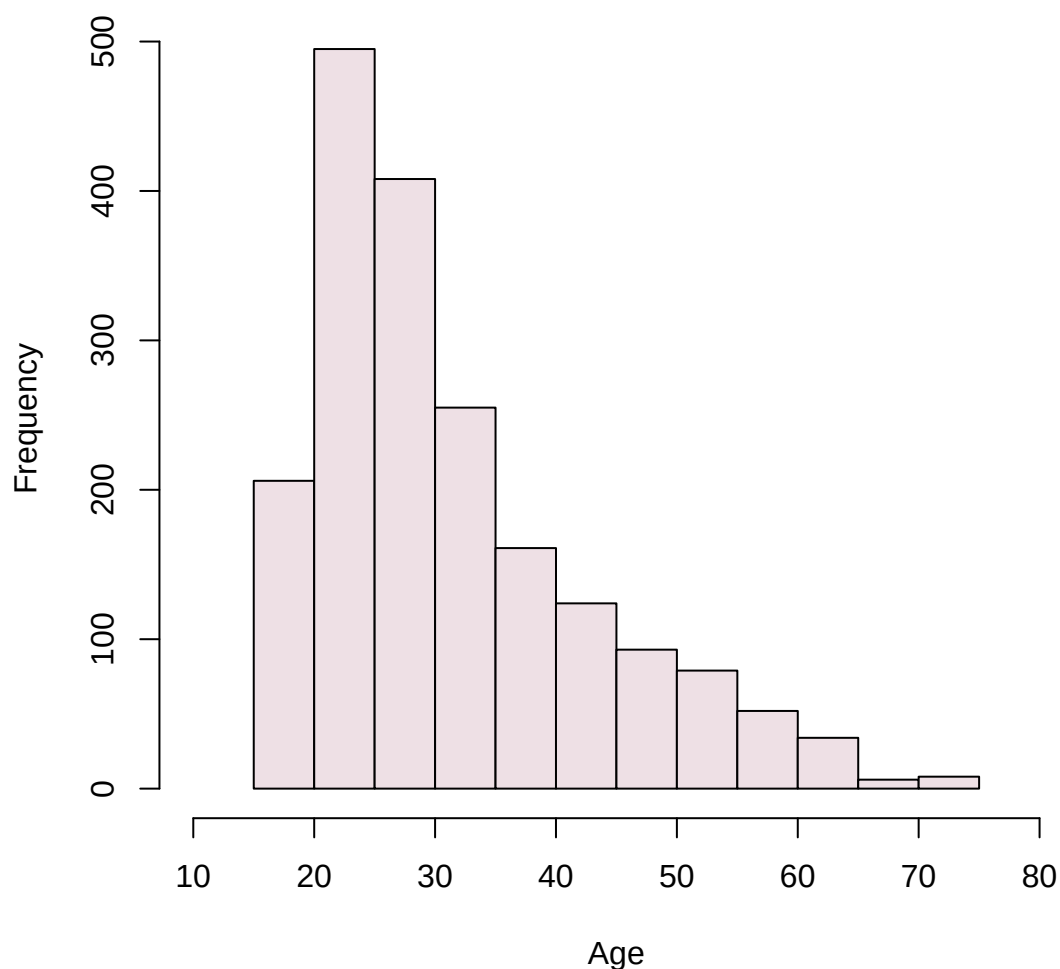


```
#creates barplot, with arguments for formatting
```

Make a histogram of the age variable.

```
hist(mturkstress$age, col="lavenderblush2", xlab="Age", xlim=c(10,80), main= "Histogram of Respondent Age")
```

## Histogram of Respondent Age

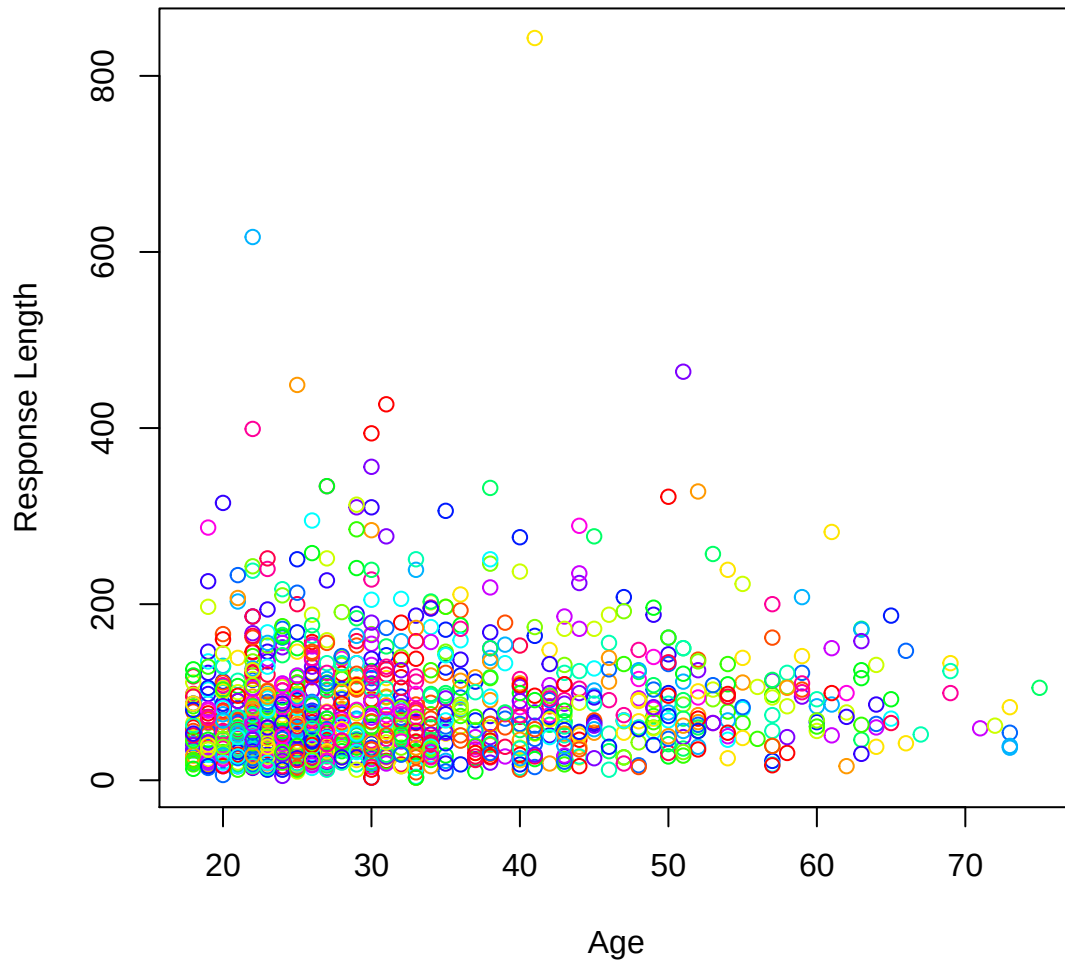


```
#creates histogram, additional arguments for formatting
```

Make a correlation plot between the age variable and the response length variable, for all those people whose response is greater than 0. What is the correlation between those two variables (you will need to run a correlation test to get the answer to this)?

```
x <- mturkstress$Age[which(mturkstress$responselength > 0)]  
#creates variable x for various ages of respondents when their response length was longer than 0 (when t  
  
y <- mturkstress$responselength[which(mturkstress$responselength > 0)]  
#creates variable y for length of responses as long as they were over 0 (so matches in number with x)  
  
plot(x,y, col=rainbow(20), xlab="Age", ylab="Response Length", main="Response Length by Age")
```

## Response Length by Age



```
#creates plot of all response length against age
```

```
cor(x,y)
```

```
## [1] 0.1202347
```

```
#runs correlation test to get level of correlation
```

The correlation variable for response length and age is .1202.